

Tutorial Sheet 3 (Double Integrals, Change of Order, Change of Variable, Beta-Gamma Functions)

1. Sketch the region of integration and evaluate the integrals:

a. $\int_{y=1}^{\ln 8} \int_{x=0}^{\ln y} e^{x+y} dx dy$

c. $\int_{x=0}^{\pi} \int_{y=0}^{\sin x} y dy dx$

b. $\int_{y=0}^2 \int_{x=0}^{y^2} e^{\frac{x}{y}} dx dy$

d. $\int_{y=-1}^0 \int_{x=-1}^1 (x + y + 1) dx dy$

2. Evaluate $\int \int_S xy - y^2 dx dy$ where S is the triangle with vertices $(0,0)$, $(10,1)$ and $(1,1)$.

3. Evaluate $\int \int_R y dx dy$, where R is the region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$.

4. Change the order of integration and evaluate the following:

a. $\int_{x=0}^{\infty} \int_{y=x}^{\infty} \frac{e^{-y}}{y} dy dx$

c. $\int_{x=0}^{4a} \int_{y=\frac{x^2}{4a}}^{2\sqrt{ax}} dy dx$

b. $\int_{x=0}^{\infty} \int_{y=0}^x x e^{\frac{-x^2}{y}} dy dx$

d. $\int_{x=0}^{\sqrt{\pi}} \int_{y=x}^{\sqrt{\pi}} \cos y^2 dy dx$

5. Sketch the region of integration and evaluate $\int \int_R (y - 2x^2) dx dy$ where R is the region inside the square $|x| + |y| = 1$.

6. Evaluate $\int \int_R e^{x^2+y^2} dy dx$ where R is the semicircle region bounded by the x -axis and the curve $y = \sqrt{1-x^2}$.

7. Evaluate the following by changing into polar coordinates:

a. $\int_{x=2}^2 \int_{y=0}^{\sqrt{2x-x^2}} \frac{x}{\sqrt{x^2+y^2}} dy dx$

b. $\int_{y=0}^2 \int_{x=0}^{\sqrt{2^2-y^2}} (x^2 + y^2) dy dx$

8. Express the following integrals as β -function:

a. $\int_0^1 x^4 (1 - \sqrt{x})^5 dx$

c. $\int_0^1 \frac{x^2}{\sqrt{1-x^5}} dx$

b. $\int_0^1 x^2 (1 - x^3)^{\frac{3}{2}} dx$

9. Evaluate the following integrals using an appropriate substitution:

a. $\int_0^{\infty} \sqrt{x} e^{-\sqrt[3]{x}} dx$

b. $\int_0^1 \frac{1}{\sqrt{-\log x}} dx$

10. Express the following integrals in terms of the Beta or Gamma function:

a. $\int_0^{\frac{\pi}{2}} \sin^3 x \sin^{\frac{5}{2}} x dx$

c. $\int_0^1 x^2 (1 - x^3)^{\frac{3}{2}} dx$

b. $\int_0^{\frac{\pi}{2}} \sin^5 x dx$

11. Prove that

$$\int_0^{\frac{\pi}{2}} \sqrt{\sin \theta} d\theta \cdot \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin \theta}} d\theta = \pi$$

12. Show that

$$\int_0^{\frac{\pi}{2}} \sqrt{\tan x} dx = \frac{1}{2} \Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{4}\right)$$

Answers:

1. a. $8 \ln 8 - 16 + e$ b. $e^2 - 1$ c. $\frac{\pi}{4}$ d. 1

2. 6

3. $\frac{48}{5}$

4. a. 1 b. $\frac{16}{3} a^2$ c. $\frac{1}{2}$ d. 0

5. $\frac{-2}{3}$

6. $\frac{\pi(e-1)}{2}$

7. a. $\frac{4}{3}$ b. 2π

8. a. $2\beta(10,6)$ b. $\frac{1}{3}\beta(1, \frac{5}{2})$ c. $\frac{1}{5}\beta(\frac{3}{5}, \frac{1}{2})$

9. a. $\frac{315}{16}\sqrt{\pi}$ b. $\sqrt{\pi}$

10. a. $\frac{8}{77}$ b. $\frac{8}{15}$ c. $\frac{\sqrt{\pi}}{2}$